A sticky elastic rod against a rigid wall

Surviving-bond traction:
rationale, equations, and implementation

`mfemMechanics` / Cu–Cu hybrid-bonding experiments

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Abstract

An elastic rod, an exact rigid wall, and a nonfailing adhesive spring isolate one question: how does pressure-assisted bond formation affect subsequent pulling and separation? A single active fraction, β, controls both loss and recovery of adhesion. Rupture responds to the traction carried by the surviving bonded portion, while the measured pulling stress is averaged over the full interface. This distinction lets rupture continue during an activated open hold even as the measured force falls. The note derives that feedback, the bounded implicit update and its time-step safeguard, and the protocols used in the interactive notebook. The model is an illustrative, isothermal, quasistatic experiment, not a calibrated law for copper.

1 A small model with an explicit physical interpretation

The motivating HBM problem involves thermo-inelastic contact, evolving microstructure, and three-dimensional constraints. The design discussion deliberately reduced that problem to one rod against a rigid wall so that contact-enforcement errors and bulk constitutive effects would not obscure the bonding law [1]. Exact contact is therefore a means of isolating the mechanism, not a proposed approximation for every Cu–Cu joint.

The spring is linear at fixed β and has no independent failure law. Instead, $\beta \in [0, 1]$ is the *current fraction of active bonds*: compression can increase it and rupture can decrease it. It is not cumulative irreversible damage. There is no second damage variable, maximum-opening history, or activation latch. Even when $\beta = 0$, ordinary contact must still prevent penetration.

The choice of rupture driver is consequential. A purely opening-velocity driver stops acting when motion stops, even if bonds remain stretched. A switch based on current nominal traction can also stop rupture during relaxation, because that traction decreases as bonds disappear. Permanently latching that switch avoids arrest but imposes continued decay even after deliberate unloading. The present choice instead asks how strongly the *surviving bonds* are loaded. Under the assumed uniform parallel-bond geometry, this produces continuing rupture during an activated held opening but permits unloading to stop further loss.

This interpretation is an assumption, not validation. Equal load sharing among closed bonds has a clear example in Erdmann and Schwarz's biological adhesion-cluster model: their Fig. 1 distributes a constant total load among i closed bonds and gives an individual off-rate $k = k_0 \exp(f/i)$, with dimensionless total force f [2]. That source illustrates load sharing, not copper physics or the rate function adopted here. The exact rod reduction and numerical derivatives below are derived from the stated equations. The bounded smooth switch is a phenomenological adaptation of the earlier template, not a formula taken from that paper.

2 Rod, wall, and the two traction measures

2.1 Geometry and equilibrium

The reference rod occupies $x \in [-L, 0]$, has area $A > 0$ and modulus $E > 0$, and faces a wall at spatial coordinate zero. The prescribed left displacement is $U(t)$ and the right-tip displacement is $u(t)$. Positive motion is toward the wall; the normal gap is $g = -u \geq 0$. Normal refers to the interface orientation, not the vertical axis of a drawing. The reference interface area is assumed to equal the rod area and remains fixed.

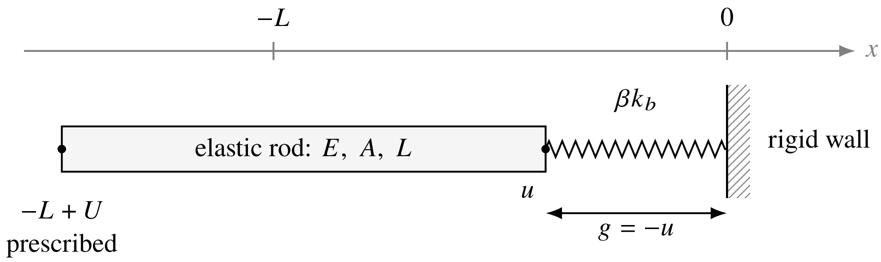

Figure 1: An opened configuration, not to scale. A positive gap is compatible with surviving adhesion. Compressive contact acts at zero gap and is not scaled by the active bond fraction.

The rod is homogeneous, small-strain, linear elastic, and uniaxial, not a plane-strain reduction. Body forces, inertia, creep, plasticity, friction, tangential motion, diffusion, and temperature evolution are omitted. With displacement field w, its strain and tension-positive force are

$$\varepsilon = w_{,x} = \frac{u - U}{L}, \qquad N = EA\varepsilon = k_r(u - U), \qquad k_r = \frac{EA}{L}. \tag{1}$$

Let $K_n \geq 0$ be fully bonded interface stiffness per area, with stress per length units, and $k_b = AK_n$ the corresponding force per length stiffness. The attractive spring force is $F_b = \beta k_b g$. The wall's compressive reaction has magnitude P and acts in the opposite direction:

$$N = F_b - P, \qquad P \geq 0, \quad g \geq 0, \quad Pg = 0. \tag{2}$$

The actuator force on the left node is $-N$; the code reports N.

For virtual displacements with $\delta w(-L) = 0$, the weak equation is

$$\int_{-L}^{0} EAw_{,x}\delta w_{,x}\,\mathrm{d}x - (F_b - P)\delta w(0) = 0. \tag{3}$$

Linear shape functions give the stiffness $k_r \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ and the reduced residual $[k_r(u - U) - F_b + P]\delta u = 0$. One-point quadrature integrates the constant rod stiffness exactly. With no body force, this linear element also represents the exact axial equilibrium displacement.

For $U > 0$, contact gives $g = u = F_b = 0$ and $P = k_r U$. For $U < 0$, a contacting tip would require a negative reaction, so $P = 0$ and $(k_r + \beta k_b)g = -k_r U$. Both branches are

$$\boxed{P = k_r \langle U \rangle_+, \quad g = \frac{\langle -U \rangle_+}{1 + r\beta}, \quad u = -g, \quad F_b = \beta k_b g, \quad N = F_b - P, \quad r = \frac{k_b}{k_r}.} \tag{4}$$

Here $\langle z \rangle_+ = \max(z, 0)$. The denominator is positive on the admissible domain. No penalty, multiplier iteration, or numerical contact stiffness is introduced; the physical contact reaction has been condensed analytically.

2.2 Measured stress versus surviving-bond loading

Pressure and nominal tensile traction use the full area, while surviving-bond traction uses the active portion under the uniform load-sharing assumption:

$$p = \frac{P}{A}, \qquad \sigma = \frac{F_b}{A} = \langle N/A \rangle_+ = \beta K_n g, \qquad \tau = \frac{\sigma}{\beta} = K_n g \quad (\beta > 0). \tag{5}$$

The code calls these `pressure`, `traction`, and `bond_traction`, respectively. It evaluates $\tau = K_n g$ directly, never by dividing by a small β. At $\beta = 0$ this expression is a limiting trial demand, not an actual stress on nonexistent bonds: nominal force and bond loss are zero. A diagnostic can omit such states from a logarithmic plot without changing the stored arrays or imposing a failure cutoff.

This choice retains a linear spring and a single history variable. It is mathematically a stress-scaled gap driver, since K_n is constant; changing the terminology alone does not add microscopic physics. The meaningful change from a fixed nominal-stress switch is that the decreasing active fraction no longer reduces the quantity used to activate rupture.

3 Formation, rupture, and unloading

Physical time remains essential despite quasistatic mechanical equilibrium. The competing rates are

$$\dot{\beta} = a(1-\beta) - b\beta, \qquad a = k_f \frac{p}{p_{\mathrm{ref}}}, \qquad b = k_d S_b(\tau). \tag{6}$$

Both rate constants are nonnegative and have inverse-time units, and $p_{\mathrm{ref}} > 0$. The switch is

$$S_b(\tau) = \begin{cases} 0, & \tau \le \tau_0, \\ 3z^2 - 2z^3, & \tau_0 < \tau < \tau_0 + \Delta\tau, \\ 1, & \tau \ge \tau_0 + \Delta\tau, \end{cases} \qquad z = \frac{\tau - \tau_0}{\Delta\tau}. \tag{7}$$

The parameters `bond_sigma0` and `bond_delta_sigma` denote $\tau_0 \ge 0$ and $\Delta\tau > 0$. They are *bonded-area stress scales*, not nominal strength values. The width is a constitutive parameter, independent of the time-step size. The switch and its first derivative are continuous. There is no additional power multiplier, baseline rupture rate, permanent activation flag, or independent cohesive damage law.

In contact $g = \tau = 0$, so only formation acts. In opening $p = 0$, so only rupture can act. At $\beta = 0$, the rate is $a \ge 0$; at $\beta = 1$, it is $-b \le 0$. These signs preserve the continuous interval $[0, 1]$. The existence of a gap does not alone imply loss: its bonded-area demand must exceed the onset threshold.

3.1 Why falling nominal traction does not stop an activated hold

Fix the actuator at $U = -G < 0$. From Eq. (4),

$$g = \frac{G}{1 + r\beta}, \quad \sigma = \frac{\beta K_n G}{1 + r\beta}, \quad \tau = \frac{K_n G}{1 + r\beta}, \quad \frac{\mathrm{d}\sigma}{\mathrm{d}\beta} = \frac{K_n G}{(1 + r\beta)^2}, \quad \frac{\mathrm{d}\tau}{\mathrm{d}\beta} = -\frac{r K_n G}{(1 + r\beta)^2}. \tag{8}$$

As bonds disappear, the gap grows and the rod unloads. The measured nominal traction decreases, but surviving-bond traction increases. If the switch was already active, it cannot become inactive during that fixed-actuator hold. Once fully activated it remains fully activated, without storing an extra flag. Then the continuous solution is

$$\beta(t) = \beta(t_*) \exp[-k_d(t - t_*)]. \tag{9}$$

More generally, an initially positive switch value bounds the subsequent loss rate away from zero during the hold, so β, nominal force, and rod strain approach zero when $k_d > 0$. A positive fraction does not reach exactly zero in finite time at bounded rates, apart from eventual floating-point underflow. There is no positive arrest fraction caused by falling nominal stress.

This does not mean every experiment must debond. A subthreshold held state is stationary. Deliberately decreasing G can unload surviving bonds and stop further loss; existing loss is not undone. After compressive recontact, formation can rebuild β. This distinction is why the model does not permanently latch the switch. It is also why merely looping an animation is not equivalent to simulating additional physical loading cycles.

3.2 Pressure exposure, strength, and energy

During compression, the exact continuous formation result is

$$\beta_{\text{end}} = 1 - (1 - \beta_{\text{start}}) \exp\left[-\frac{k_f}{p_{\text{ref}}} \int p(t)\,\mathrm{d}t\right]. \tag{10}$$

Loading and unloading ramps contribute alongside the pressure hold. Thus the baseline combines pressure and time through integrated exposure rather than two independent microscopic mechanisms. Formation identifies the ratio k_f/p_{ref} unless one factor is specified independently. Before opening loss starts, an active fraction β_b gives nominal onset $\sigma_{\text{onset}} = \beta_b \tau_0$. The nominal onset therefore scales with the amount of bonding; it is not a fixed threshold independent of the pressure/hold history. The later force peak also depends on loading speed, compliance, and kinetics, so onset is not a hard maximum-strength cap.

The elastic interface energy per nominal area provides a useful sign check:

$$\psi_{\text{int}} = \tfrac{1}{2}\beta K_n g^2, \qquad \partial_g \psi_{\text{int}} = \sigma, \qquad \mathcal{D}_{\text{loss}} = -\tfrac{1}{2} K_n g^2 \dot{\beta} \geq 0. \tag{11}$$

Formation acts at $g = 0$ and therefore adds no elastic spring energy in this idealization. This is not a chemical thermodynamic proof: chemical free energy, diffusion, and temperature dependence have not been specified. The law also does not prescribe a calibrated fracture energy. For quantitative copper predictions, full load histories, work of separation, failure location, and possible bulk inelasticity need experimental assessment.

4 The experiment and its parameters

Every cycle begins at $U = -D$ and ends at the same actuator coordinate. In contact the target displacement is $U_{\text{press}} = p_{\text{peak}} L/E$. The pull duration is D/v_{pull}, independent of peak pressure. Separating compression unloading from tensile pulling avoids inadvertently changing tensile speed when pressure is varied.

Table 1: The piecewise-linear loading protocol; zero-duration holds are omitted.

Phase	Duration	Prescribed motion or hold
Approach / return	t_{approach}	$-D \longrightarrow 0$
Press	t_{press}	$0 \longrightarrow U_{\text{press}}$
Compressed hold	t_{hold}	$U = U_{\text{press}}$
Unload compression	t_{unload}	$U_{\text{press}} \longrightarrow 0$
Tensile pull	D/v_{pull}	$0 \longrightarrow -D$
Open-actuator hold	$t_{\text{open hold}}$	$U = -D$

The next approach is a return of the same interface. Its surviving bonds can still rupture while sufficiently stretched; pressure-driven formation begins only after contact is restored. No state is reset at the cycle boundary. Conversely, each *independent sweep case* starts from the same specified β_0. Its pressure dose per cycle is

$$Q_p = p_{\text{peak}} \left(t_{\text{press}}/2 + t_{\text{hold}} + t_{\text{unload}}/2 \right) . \tag{12}$$

Equation (10) applies to the compression interval starting with the fraction after approach, not to an entire cycle that also includes rupture.

Table 2: Module defaults. All units are internally consistent but arbitrary; no value is a measured copper property. Stress is force per area.

Input	Default	Meaning / units
`E`, `A`, `L`	100, 1, 1	Modulus [stress], area, length
`K_n`	10	Fully bonded stiffness [stress/length]
`k_f`, `p_ref`	1, 0.1	Formation rate [1/time], pressure scale [stress]
`k_d`	1	Fully activated rupture rate [1/time]
`bond_sigma0`	0.04	Bonded-area onset τ_0 [stress]
`bond_delta_sigma`	0.02	Bonded-area activation width $\Delta\tau$ [stress]
`beta0`, `p_peak`	0, 0.1	Initial fraction, peak pressure [stress]
`t_approach`, `t_press`	1, 1	Ramp durations [time]
`t_hold`, `t_unload`	3, 1	Compressed hold and unload durations [time]
`pull_distance`, `v_pull`	0.12, 0.04	Retraction [length], speed [length/time]
`t_open_hold`, `cycles`	3, 1	Open hold [time], number of physical cycles
`dt`	0.03	Maximum internal step and output spacing [time]
`rtol`, `atol`	$10^{-5}, 10^{-9}$	Relative and absolute beta error scales

All inputs and derived scales must be finite. Modulus, area, length, pressure scale, activation width, non-hold durations, pull distance and speed, and numerical step/tolerance scales are positive. Rates, interface stiffness, onset, peak pressure, and hold durations may be zero but not negative. The fraction lies in $[0, 1]$ and the cycle count is a positive integer. The code rejects $p_{\text{peak}} \geq E$ to avoid rod inversion, but the actual small-strain requirement is much stronger: $|N|/(EA) \ll 1$ throughout the history. A large actuator travel can mainly become gap rather than rod strain. Area scaling is explicit: changing A scales forces and both total stiffnesses together, leaving the stress-driven beta trajectory unchanged.

The notebook's *Visible Stickiness* preset uses $E = 5$, $K_n = 250$, $p_{\text{peak}} = p_{\text{ref}} = 0.05$, $\tau_0 = 0.02$, $\Delta\tau = 0.01$, $k_d = 2$, $D = 0.08$, and $v_{\text{pull}} = 0.015$. The compressed and open holds are 4 and 8 time units; it uses one cycle and `dt` $= 0.01$, with other inputs unchanged. These explicitly bonded-area thresholds are not a conversion of earlier nominal-strength data. The stiffness ratio is 50: with beta near one, only about $1/51$ of initial retraction moves the tip, making elastic stretching easier to see. The default pressure sweep ends at 0.1, limiting its compression strain to two percent.

5 Implicit integration without spurious rupture

5.1 Residual, derivative, and the uniqueness safeguard

After mechanical condensation, only beta requires integration. Backward Euler at a prescribed endpoint U_{n+1} gives

$$R(\beta) = \beta - \beta_n - ha(1 - \beta) + hb(\tau(\beta))\beta = 0. \tag{13}$$

In compression $b = 0$ and the exact solution of this *discrete* equation is $(\beta_n + ha)/(1 + ha)$. This is not the exact continuous formation exponential. In opening, $a = 0$, and for $G = -U_{n+1} > 0$,

$$R'(\beta) = 1 + hb - h\beta b_\tau \frac{r\tau}{1 + r\beta}, \qquad b_\tau = \frac{6k_d z(1 - z)}{\Delta\tau} \quad \text{inside the switch interval.} \tag{14}$$

Outside the interval $b_\tau = 0$. The negative term is the mechanical feedback of increasing surviving-bond demand. Unlike a nominal-stress gate, a large opening step can therefore make the residual nonmonotone.

The code first handles stationary and fully activated branches. If the old fraction evaluated at the new actuator position is subthreshold, it retains that stationary branch instead of selecting a spurious large-step damaged root. If it is fully activated, every smaller fraction is also fully active, and the unique update is $\beta_n/(1 + hk_d)$. A zero old fraction remains zero during opening without division. These are exact discrete branch solutions.

For a partially activated old state, define $T = K_n G$ and

$$B = \frac{r\beta_n}{1 + r\beta_n} \min(T, \tau_0 + \Delta\tau) \frac{1.5k_d}{\Delta\tau}. \tag{15}$$

This bounds $\beta b_\tau |\mathrm{d}\tau/\mathrm{d}\beta|$ throughout $[0, \beta_n]$: the fractional prefactor increases with beta, the switch slope is at most $1.5k_d/\Delta\tau$, and wherever that slope is nonzero the traction is at most $\min(T, \tau_0 + \Delta\tau)$. Requiring $hB \le 0.8$ gives $R' \ge 0.2$. Since $R(0) = -\beta_n$ and $R(\beta_n) \ge 0$, this guarantees one root in the admissible decay bracket. The factor 0.8 is a dimensionless solver safety margin, not material damping or a constitutive parameter.

To retain relative accuracy for tiny fractions, solve $q = \beta/\beta_n$:

$$H(q) = q[1 + hb(\tau(\beta_n q))] - 1 = 0, \qquad 0 \le q \le 1. \tag{16}$$

Its analytic derivative follows from $\tau_q = -r\beta_n\tau/(1 + r\beta_n q)$. Safeguarded Newton proposals outside the central 80 percent of the current bracket are replaced by bisection. A failed uniqueness bound or scalar solve rejects the trial; it never clips beta or commits an unconverged iterate. The continuous stationary solution and the guarded discrete root must not be confused with choosing an arbitrary root of an unrestricted large-step equation.

5.2 Time accuracy and reproducibility across cycles

Adaptive step doubling compares one full BE step with two consecutive half steps from the same accepted state, using the actuator value at each step's own endpoint. The normalized error is

$$e = \frac{|\beta_{\text{two}} - \beta_{\text{full}}|}{\texttt{atol} + \texttt{rtol}\max(\beta_n, \beta_{\text{two}})}. \tag{17}$$

If $e \le 1$, the two-half-step result is accepted; no Richardson extrapolation is used because it need not preserve bounds. Rejected trials advance neither time nor state. The scheme remains first order globally. A local error tolerance is not a guarantee on the final history or sampled peak stress.

All load corners are output events and internal steps do not cross them. Integration uses phase-local time and the exact input phase duration, so shifting the same cycle to a later absolute time does not change adaptive decisions through cancellation in endpoint subtraction. Stored output times are then shifted to the global timeline, retaining exact phase endpoints. The code validates representable output spacing and limits outputs to 200,000 points, adaptive trials to 2,000,000, and independent sweep cases to 128. Exceeding a resource or representability limit is reported, not silently shortened or regularized.

6 Reading and verifying the notebook

The frozen `Parameters` object defines one run. `load_segments` records cycle, phase, start, end, duration, and the two endpoint displacements. `simulate` returns the accepted arrays and step statistics; `summarize_cycles` reports nominal peak stress, its time, initial and final fractions, pressure dose, and equilibrium/complementarity errors. The peak is taken during tensile pulling and its following open hold, not during the next return. A shared boundary sample belongs to the preceding phase label, so summaries use inclusive time bounds rather than labels alone. The `peak_at_pull_end` flag marks a positive maximum at the prescribed pull endpoint. It is potentially range-limited, not proven pull-off strength.

`sweep` varies only peak pressure or compressed hold time, starting each case independently from the same initial fraction and carrying history within any subsequent physical cycles. Peak stress always means nominal stress, never the much larger bonded-area demand. The diagnostic panels show both measures separately. A constant bonded-area threshold must not be overlaid as a constant nominal-strength line; the equivalent nominal onset is $\beta\tau_0$ and changes with the active fraction.

The animation consumes an already computed trajectory. It interpolates U and beta in physical time, recomputes mechanics, and only then applies a display-only magnification $s \geq 1$:

$$x_{\text{left}}^{\text{display}} = -L + sU, \qquad x_{\text{tip}}^{\text{display}} = su, \qquad L_{\text{rod}}^{\text{display}} = L + s(u - U). \tag{18}$$

The default notebook factor is 10; 1 restores actual geometry. A dashed unstretched-length guide follows the actuator. It is not a second rod and may cross the wall during compression. Numeric strain, gap, pressure, and both traction labels remain physical, as do all history plots. A magnification that collapses the displayed compressed rod is rejected. Adhesive opacity tracks beta continuously, with no visual failure cutoff. The initial movie uses 120 frames at 5 fps, about 24 seconds for one cycle. The loop button replays that trajectory; changing the physical cycle count is a separate action.

Regression tests check mechanical equilibrium, contact complementarity, switch and residual derivatives by centered differences, stationary and fully activated branches, the uniqueness safeguard, tiny fractions, area scaling, and invalid inputs. In particular, an activated fixed-actuator hold must continue losing bonds while nominal traction decreases, following the exponential oracle once fully active. Closure must be able to stop loss and compression must permit formation without an activation reset. Cycle-by-cycle and continuous runs must agree; sweep values must not inherit each other's history. Time refinement checks histories and nominal peaks, not just force balance. Animation tests distinguish physical mechanics from display scaling.

These are implementation checks, not experimental validation. The notebook retains executed results, exact parameters, dependency versions, and a source hash. This document explains the assumptions and algorithms rather than exporting simulation results. The README contains setup and launch instructions.

References

[1] *Formulate Weak Form*. Shared ChatGPT conversation, design history. Accessed 6 September 2026. https://chatgpt.com/share/6a9dc146-1d1c-83e8-bbb7-e159d93ad7c8. Used for the minimal rod/contact model, the nonfailing spring requirement, and the evolution of candidate drivers; not physical validation.

[2] T. Erdmann and U. S. Schwarz, "Stability of Adhesion Clusters under Constant Force," *Physical Review Letters* **92**, 108102 (2004). doi:10.1103/PhysRevLett.92.108102. Figure 1 is used for uniform sharing among closed bonds; its biological constant-force off-rate is not adopted as the present copper interface law.